

UAV Fuselage & Landing Gear Configurator

PRODUCT MODELLING PROJECT | GROUP 22

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Team Findus

DRIVING DESIGN BY MISSION

Traditional manual layout of UAV internal components is time-consuming and prone to human error during conceptual iterations.

The Solution: A parametric, mission-driven configurator that automates internal structural generation based on payload and mission specs.

- Removes manual repositioning work
- Ensures structural consistency
- Closes the design loop for CG/Landing Gear



| STRATEGIC SYNERGY: GROUP 21 & 22



Group 21: Wing and Tail

Responsible for the high-lift wing system and empennage design. Focuses on external aerodynamic surfaces.



Group 22: Internal Integration

Our focus: The "Belly of the Beast" – internal structural ribs, landing gear, and payload optimization.

"Unified interfaces ensure wing-spar and tail-boom mounts are always compatible."

INTEGRATED SYSTEM ARCHITECTURE



VB.NET GUI

Visual Studio interface
handles user inputs, mission
profiles, and validation rules.



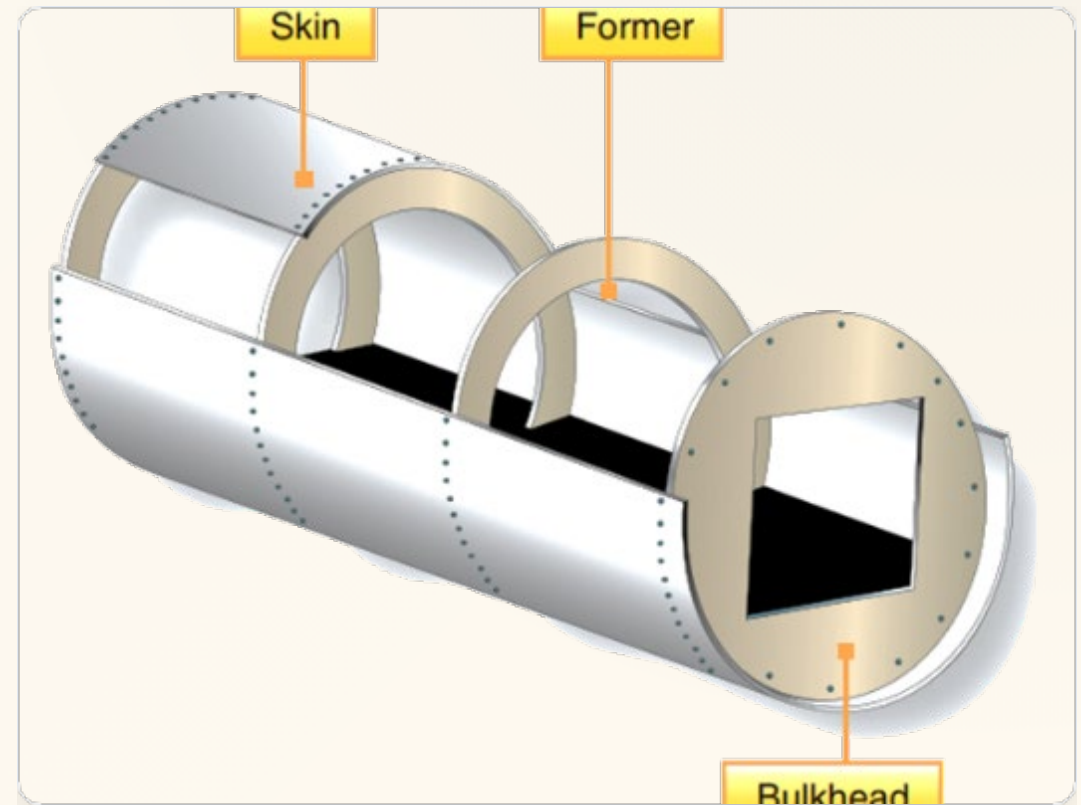
Automation Layer

CATIA V5 Interop API
translates GUI logic into 3D
geometric operations.

THE POWERCOPY STRUCTURAL LIBRARY

We built five primary reusable PowerCopies to form the semi-monocoque fuselage:

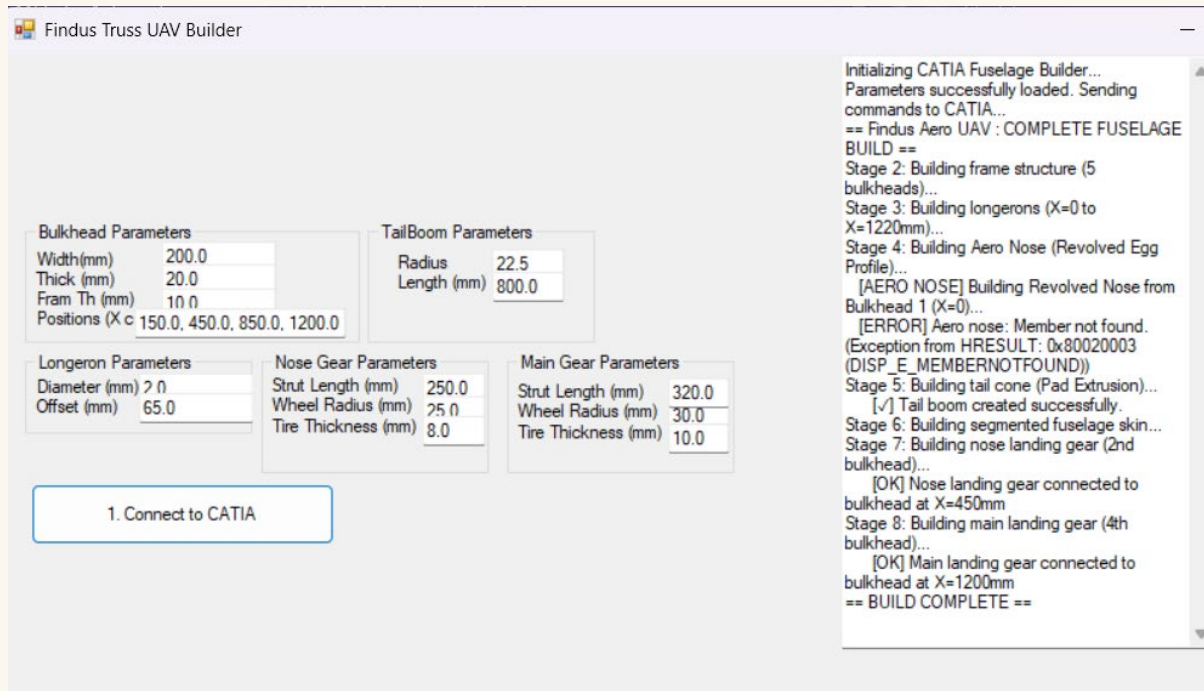
- ✓ **Bulkheads:** Key payload boundaries
- ✓ **Frames:** Light intermediate support
- ✓ **Stringers/Longerons:** Stiffeners
- ✓ **Lightening Holes:** Patterned cut-outs



MISSION PROFILE INPUTS

Category	Primary Parameters	Impact on Model
Payload	Mass (kg), Bounding Box (mm)	Fuselage Width/Height
Performance	Target Endurance, Take-off Run	Fuselage Length / Motor Class
Structure	Materials (Balsa/Plywood)	Rib Spacing & Thickness
Stability	Gear Type (Tricycle/Tail-dragger)	Wheel-base placement

AUTOMATED COMPUTATION RULES



Logic-Driven Placement

The VB.NET layer calculates the X-station of every internal item before CATIA triggers:

$$\text{Fuselage_Length} = f(\text{Payload}, \text{Battery_Volume})$$
$$\text{CG_System} = \sum(\text{Mass}_i * X_i) / \text{Total_Mass}$$
$$\text{Gear_Station_X} = \text{CG_System} + \text{Offset_K}$$

SAE COMPETITION STANDARDS



Logistic Limits

Hard-coded limits in the GUI ensure the design always fits within the SAE shipping box dimensions.



Material Presets

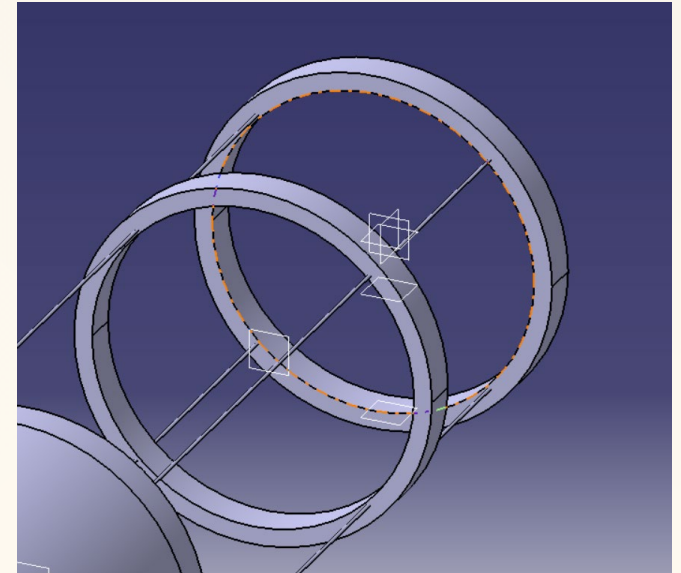
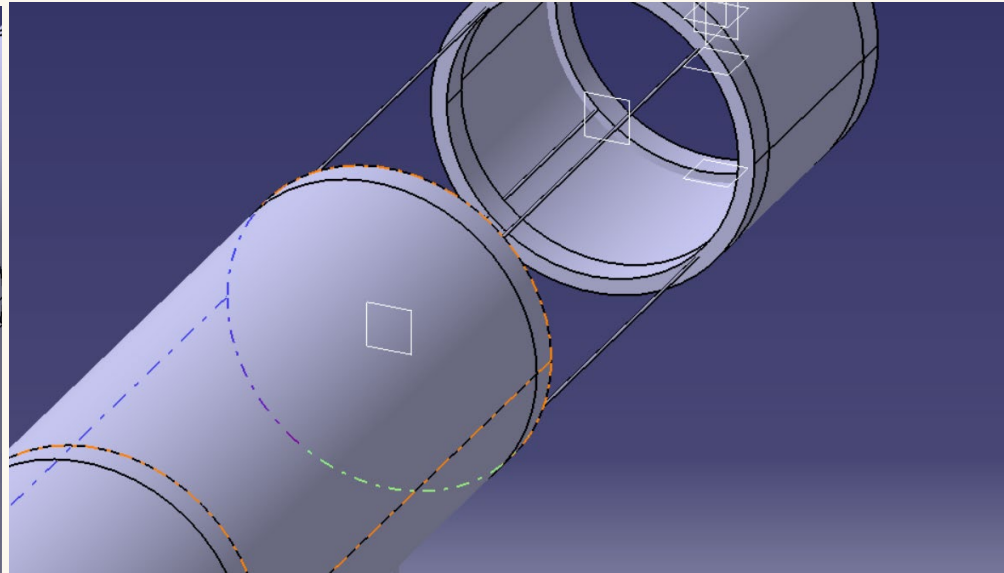
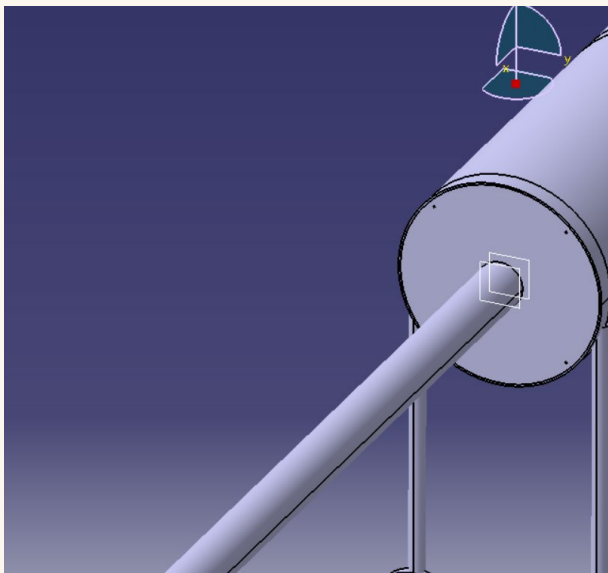
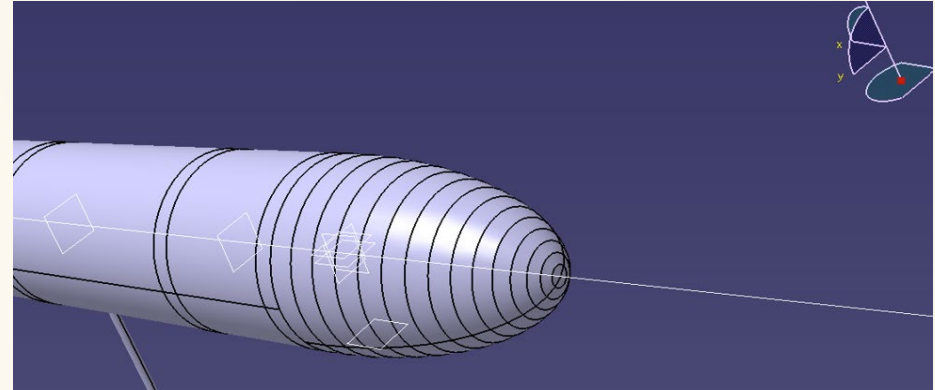
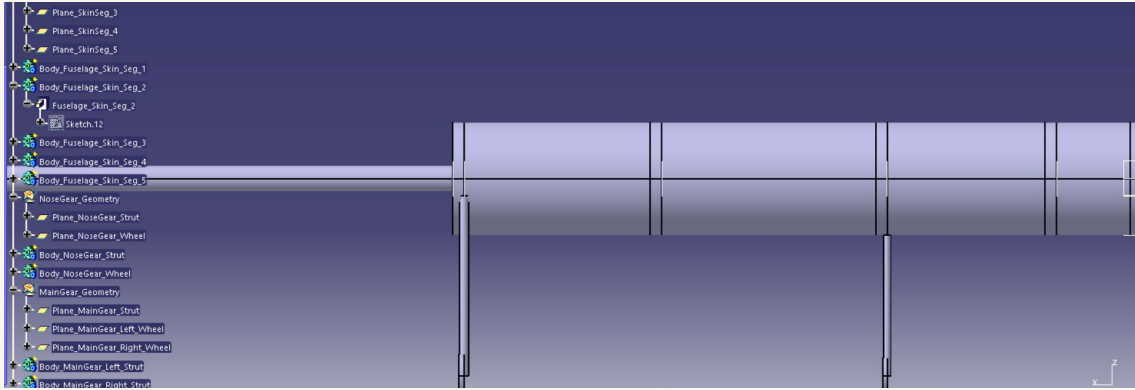
Standardized sizes for 1/8" Balsa, 3mm Plywood, and Spruce stringers for direct laser-cutting output.



Weight Efficiency

Automated lightening holes reduction patterned on frames to maximize payload capacity.

What we Achieved, A dynamic Running VB, Net Code which Generates a User Defined CAD model using our Parametric Bulkhead, Stringers, Longerons and indeed a Tail Boom Configuration for SAE



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The screenshot displays the Visual Studio Code environment with the 'Catia Automation' extension active. The main window shows a 'Findus Truss UAV Builder' dialog box with various parametric input fields for a UAV model.

Findus Truss UAV Builder Parameters:

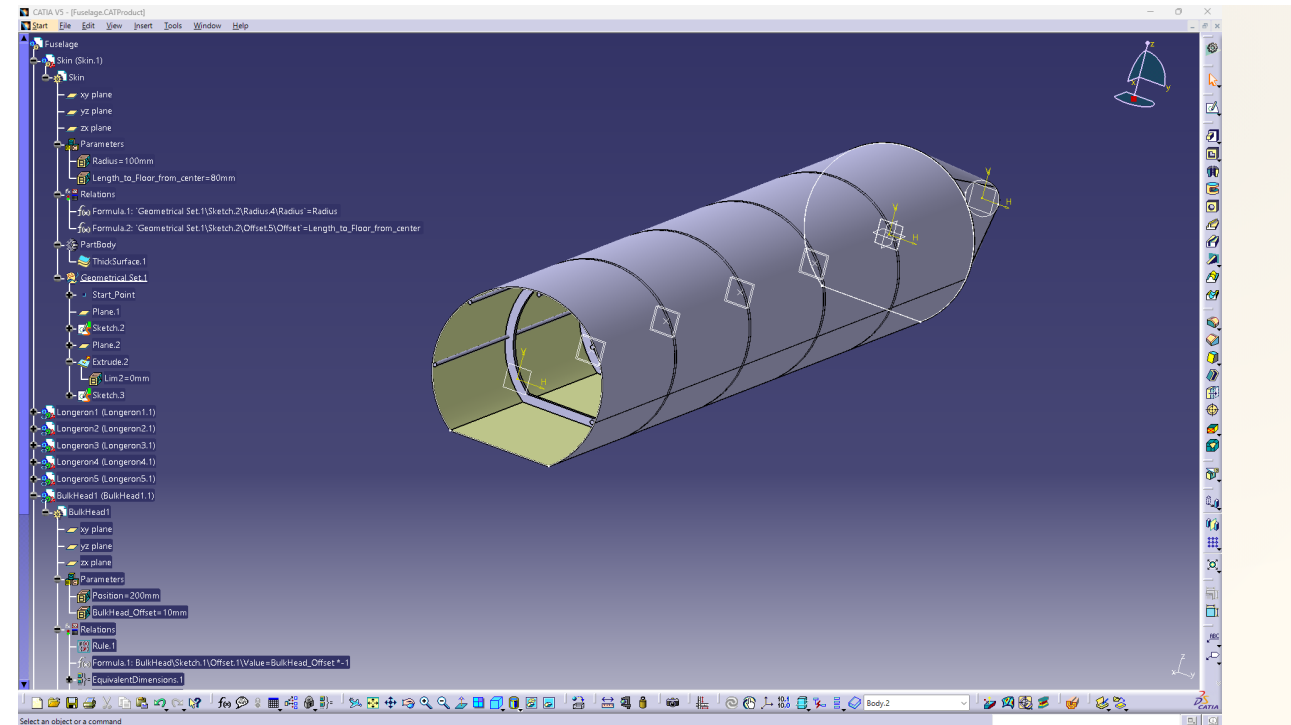
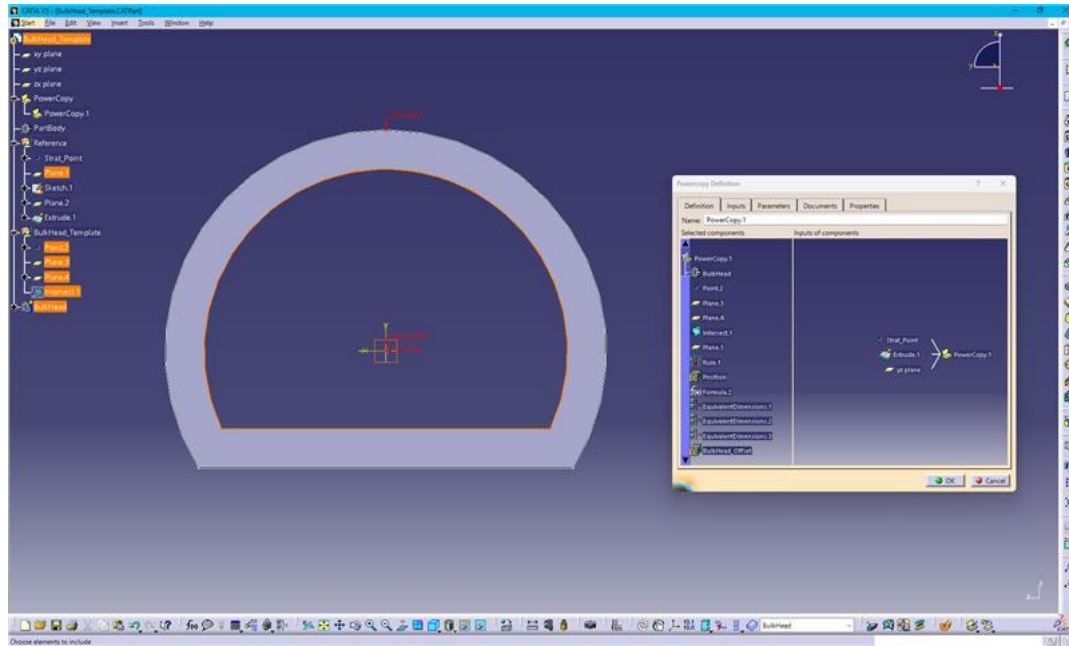
Bulkhead Parameters		Tail Boom Parameters		Longeron Parameters		Nose Gear Parameters		Main Gear Parameters	
Width (mm)	200.0	Radius	22.5	Diameter (mm)	20.0	Strut Length (mm)	250.0	Strut Length (mm)	320.0
Thick (mm)	20.0	Length (mm)	800.0	Offset (mm)	65.0	Wheel Radius (mm)	25.0	Wheel Radius (mm)	30.0
Fram Th (mm)	10.0					Tire Thickness (mm)	8.0	Tire Thickness (mm)	10.0
Positions (X c)	150.0, 450.0, 850.0, 1200.0								

A button labeled '1. Connect to CATIA' is visible at the bottom of the dialog box.

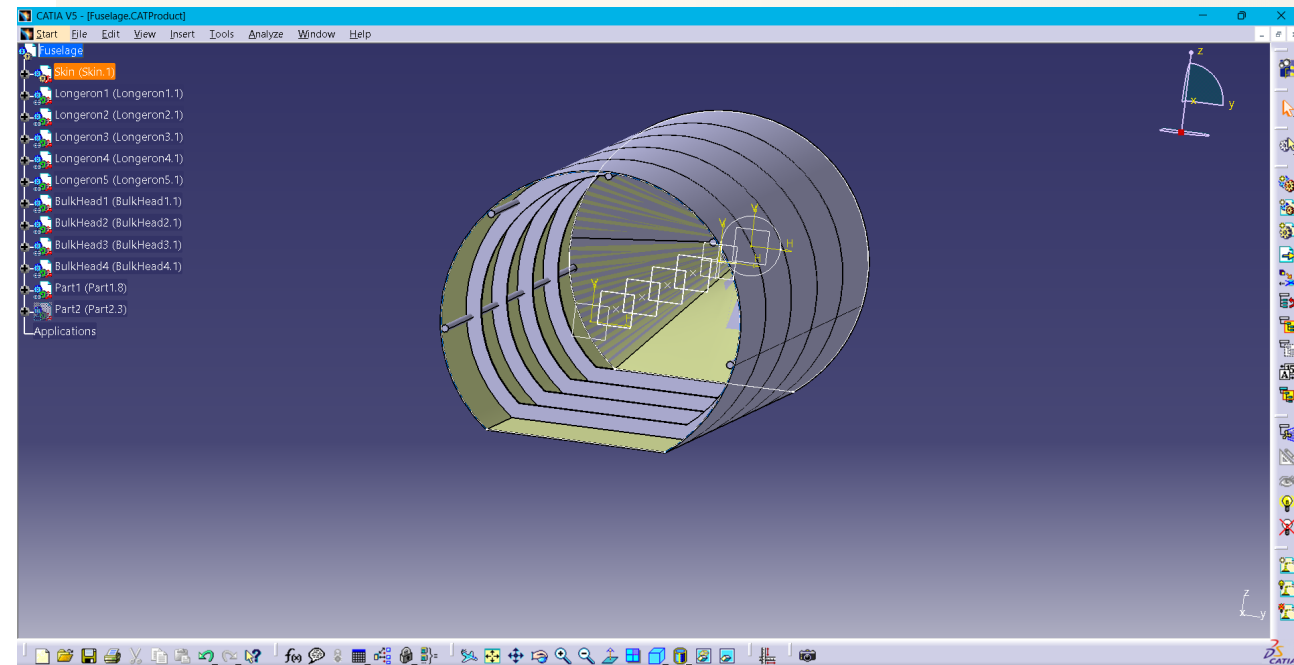
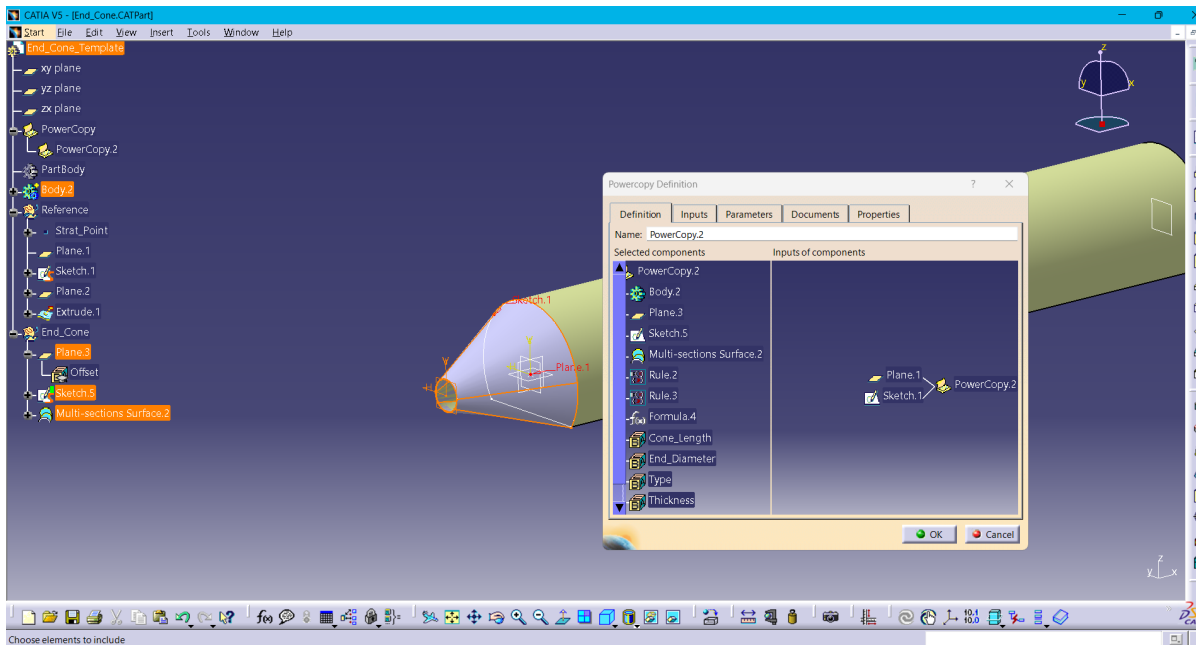
The background shows the Visual Studio Code interface with the 'Catia Automation' extension loaded. The 'Copilot Chat' sidebar is open on the right, displaying a 'New Thread' and a 'Copilot Chat' window with a message: 'I might make mistakes, so check for accuracy. Use # for references, / for commands, and @ for agents.'

The bottom status bar shows the system clock as 10:28 am on 27/05/2026.

PowerCopy Manual Instansiation → Automatic Instantiation, Bulkheads, longerons and Nose Cone Visible



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Questions & Discussion

Thank you for your attention. We are now ready to demonstrate the live CATIA configurator and GUI capabilities.

NEXT UP:

5-Minute Live Model
Demo

RESOURCES:

Teams folder (pptx, pdf,
demo.mpeg)

IMAGE SOURCES



<https://ebrary.net/htm/img/39/518/240.png>

Source: ebrary.net



<https://www.abbottaerospace.com/wp-content/uploads/2019/08/201.png>

Source: www.abbottaerospace.com



<https://cdn.mescius.io/umb/media/5udeopna/2023-mescius-c1-vb-net.png>

Source: developer.mescius.com
